

Agentic AI-Based Respiratory Disease Smart Diagnosis Assistant

AI-Powered Autonomous Diagnosis · RAG Knowledge Retrieval · Smart Appointment Scheduling

1. Problem Statement

Respiratory diseases — including Asthma, Chronic Obstructive Pulmonary Disease (COPD), Pneumonia, Tuberculosis (TB), and COVID-19 — account for more than **4 million deaths annually** worldwide. In resource-limited settings, timely and accurate diagnosis remains a critical challenge due to physician shortages, geographic barriers, and delayed specialist referrals.

Current clinical pathways rely heavily on manual symptom assessment, spirometry, and chest imaging, which are time-consuming and prone to human error. Patients often receive delayed diagnoses after multiple consultations, leading to disease progression and avoidable hospitalisations. There is an urgent need for an **intelligent, autonomous system** capable of real-time symptom triage, evidence-based disease prediction, personalised treatment guidance, and seamless appointment scheduling — all in a single, integrated platform.

■ Delayed Diagnosis	■ Limited Specialist Access	■ Knowledge Gaps
Patients undergo 3–5 consultations before an accurate respiratory diagnosis is reached.	Rural and semi-urban populations lack access to pulmonologists and diagnostic labs.	General practitioners often lack updated clinical guidelines for rare respiratory conditions.

2. Existing Solutions & Their Limitations

Rule-based symptom checkers (WebMD, Ada Health)	Static decision trees; no personalisation or medical knowledge updates.
ML-based chest X-ray classifiers (CheXNet)	Imaging-only; cannot integrate multi-modal symptom + history data.
EHR-integrated CDSSs (Epic, Cerner plugins)	Require existing electronic records; inaccessible to underserved populations.
Telemedicine platforms (Practo, DocsApp)	Human-dependent; no autonomous triage or AI-driven diagnosis.
Stand-alone NLP chatbots	Lack domain-grounded RAG retrieval; hallucination-prone and non-explainable.

3. Proposed Solution

We propose an **Agentic AI-Powered Respiratory Disease Smart Assistant** — a multi-agent, RAG-augmented system that autonomously handles end-to-end patient interactions from symptom intake to specialist scheduling. Built on a Large Language Model (LLM) backbone with retrieval-augmented generation (RAG), the assistant continuously queries an up-to-date respiratory medical knowledge base to deliver explainable, evidence-based outputs.

■ Symptom Analysis Agent	■ Disease Prediction Engine	■ RAG Medical Knowledge Base
NLP-driven intake of free-text and structured symptoms. Extracts onset, duration, severity, and comorbidities.	Multi-label classifier predicts Asthma, COPD, Pneumonia, TB, Bronchitis with confidence scores and differential diagnoses.	Vector-indexed corpus of clinical guidelines (WHO, GOLD, GINA), PubMed abstracts, and drug formularies — retrieved at inference.
■ Treatment Recommendation	■ Appointment Scheduling	■ Privacy & Explainability
Evidence-based first-line and adjunct therapy suggestions, inhaler protocols, and lifestyle modifications per patient profile.	Autonomous calendar integration to book pulmonologist or GP slots based on urgency score and geo-location.	HIPAA-compliant data handling; SHAP-based explanations for every diagnosis to build clinician and patient trust.

4. System Workflow

1	Intake Agent	Patient enters symptoms via chat/voice; NLP parser extracts structured entities.
2	RAG Retrieval Module	Embedding model encodes symptom vector; top-K relevant clinical documents fetched from vector DB.
3	Diagnosis Agent (LLM)	LLM reasons over retrieved context + patient history to generate ranked differential diagnoses.
4	Treatment Agent	Maps predicted disease + patient profile to evidence-based treatment protocols and drug recommendations.
5	Urgency Classifier	Assigns triage severity (low / moderate / critical) and triggers escalation if SpO2 or symptom threshold crossed.
6	Scheduling Agent	If urgency ≥ moderate, autonomously books nearest available pulmonologist slot and sends confirmation.
7	Explainability Layer	Generates patient-friendly summary with cited evidence, confidence levels, and action checklist.

5. Expected Outcomes & Impact

■ >90% diagnostic accuracy	■ <30 sec triage response	■ 40% reduction in unnecessary ER visits
Validated on benchmark respiratory datasets (MIMIC-III, NIH ChestX-ray14).	Real-time symptom-to-recommendation pipeline enabled by agentic LLM architecture.	Early intervention and guided home management reduce acute exacerbations.